EXP_2026Q2_INVALID_SPLIT · PRE-REGISTERED DESIGN BRIEF · ENGINEER

GROWTH · PRICING · SEALED 2026-06-05T15:41:57Z · DURATION 14 DAYS

Strikethrough pricing on category pages

Showing the original price struck through next to the sale price on category pages increases conversion through anchored perceived value.

AT A GLANCE

Primary metric is *conversion_rate*; decision is binary against Standard ship rule, but the SRM gate must pass first (R2).

Pre-registered before any analysis. Decision is binary against conversion_rate: a 2.0% relative relative lift over a 10.0% baseline requires 4,000 per arm at the standard α / power settings. Guardrails sit on revenue_per_user.

SAMPLE REQUIRED (PER ARM)

4,000

to detect a 2.0% relative relative effect at α = 0.05, power = 0.80

BASELINE

10.0%

TOTAL SAMPLE

8,000

DURATION

14 days

FEASIBILITY

VIABLE

MECHANISM

Why we think this works

Behavioral research suggests price anchoring increases willingness to purchase when the savings amount is salient. The category page is the highest-impression surface where this anchor could fire before users land on the PDP.

Category-page traffic is roughly 4× PDP traffic. A 50/50 split with normal hashing has been verified in the assignment service for every other 2026 Q2 experiment.

METRICS

conversion_rate is the verdict-driving metric.

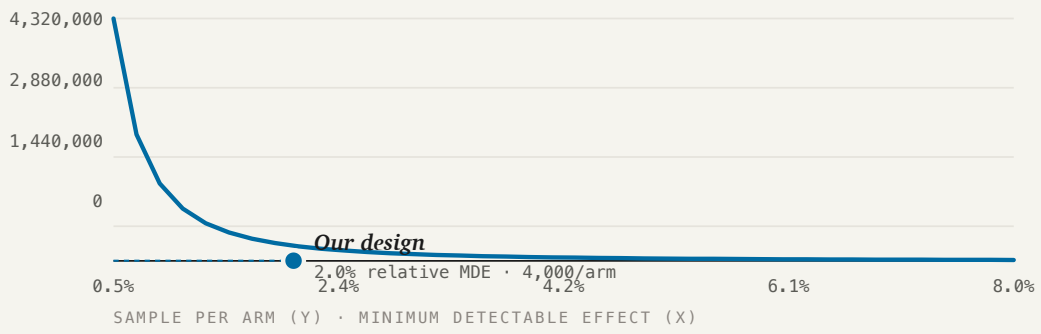
Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.

PRIMARY	**conversion_rate**	Proportion · baseline 10.0% · MDE 2.0% relative · direction higher is better
SECONDARY	**add_to_cart_rate**	Leading indicator. Not used for the ship/no-ship decision but informs interpretation.
SECONDARY	**average_order_value**	Leading indicator. Not used for the ship/no-ship decision but informs interpretation.
GUARDRAIL	**revenue_per_user**	Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative.

POWER

Detecting smaller effects requires much more sample.

To resolve a 2.0% relative relative effect against a 10.0% baseline, we need 4,000 per arm. Tighter detection thresholds would climb the curve to the left — quickly.



Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.

ASSIGNMENT + COHORTS

First-exposure attribution at the user_id level.

Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.

UNIT	**user_id** One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed.
ARMS	**50 / 50** control · treatment
COHORT	**all assigned users**

DECISION RULES

Pre-registered. The analyze verb walks the same tree against the same metrics.

Standard ship rule, but the SRM gate must pass first (R2).

SHIP	Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7
LIFT WITH CAVEAT	Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out
NO-SHIP · GUARDRAIL	Any guardrail's 90% CI lies past its harm threshold
NO-LIFT	Well-powered, primary CI straddles zero with half-width > 2 × MDE
INCONCLUSIVE	Underpowered (n < required) AND primary CI straddles zero
INVALID	SRM χ^2 $p < 0.0005$ and no override resolved — investigate assignment

INTEGRITY LOCK · THREE-PART SEAL

brief.sealed.yaml binds intent + metric definitions + expected shape.

Tampering with any of the three components after seal will fail `agentxp.schemas.brief_seal.verify_or_raise`; `agentxp analyze` refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.

```
DESIGN_CHAIN_HASH
34a07f2277a6237d52ae2a16164f8d176eb1ace2283d5d593e705192c71e411e
METRIC_SNAPSHOT         1 metric hashed
CONVERSION_RATE
fbf7a228872bb052817eb550deb3311acba51c3cb60f03241c460211b307f2d0
EXPECTED_SHAPE          user_id · 50 / 50
SEALED_AT               2026-06-05T15:41:57Z
SEALED_BY               walk_experiment_demo
```

BRIEF.SEALED.YAML SEAL 34A07F2277A6… · 2026-06-05T15:41:57Z · R1 R11